IoT-Based Indoor Environmental Safety Monitoring Using Sensor Data and Machine Learning

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Abstract

Indoor environmental conditions have a significant impact on human health and safety. Poor air quality, gas leakage, and abnormal environmental changes can lead to dangerous situations if they are not detected early. With the rapid development of Internet of Things (IoT) technologies, low-cost sensors can now be used to build intelligent monitoring systems capable of analyzing environmental conditions in real time. This research proposes an IoT-based indoor environmental monitoring system designed to improve safety within enclosed environments. The system integrates multiple sensors including the DHT22 sensor for temperature and humidity monitoring, MQ-135 gas sensor for detecting harmful gases, and the Sharp GP2Y1010AU0F optical dust sensor for monitoring airborne particulate matter. In addition, PM2.5 measurements are used to estimate indoor air quality levels. The collected environmental data is analyzed using machine learning models including Logistic Regression for classification tasks and Isolation Forest for anomaly detection. The system is capable of detecting human presence, gas leakage, potential fire hazards, and unsafe air quality conditions. External air quality information obtained through the OpenAQ API further enhances the system's monitoring capability. The proposed system demonstrates how low-cost IoT sensors combined with machine learning techniques can provide a reliable and affordable environmental safety monitoring solution.

1. Introduction

Environmental monitoring has become increasingly important due to rising pollution levels and growing concerns regarding public safety. Indoor spaces such as laboratories, storage rooms, offices, and homes can experience hazardous conditions including gas leakage, poor air circulation, and abnormal temperature changes. Without proper monitoring systems, these hazards may go unnoticed and can lead to serious health and safety risks. Traditional environmental monitoring systems often rely on expensive industrial sensors or large outdoor monitoring networks. These systems are difficult to deploy in small indoor environments and may not be economically feasible for many applications. Recent advancements in IoT technology have enabled the development of compact environmental monitoring systems using affordable sensors and microcontrollers. In this project, an IoT-based monitoring system is designed to analyze environmental conditions inside a controlled indoor environment. Small sensors tend to produce noisy and inconsistent readings

when placed outdoors due to weather conditions and environmental disturbances. Therefore, this system focuses on indoor deployment where environmental conditions are relatively stable. The system collects environmental data continuously and analyzes it using machine learning techniques to detect patterns that may indicate hazardous situations. By combining sensor data analysis with external environmental data sources such as the OpenAQ API, the system aims to provide a reliable environmental safety monitoring solution.

2. Methodology and Sensor Design

The proposed monitoring system integrates multiple sensors to measure environmental parameters. **DHT22 Sensor:** This sensor measures temperature and humidity. Humidity levels can help identify human presence in a closed room because respiration and body heat cause small increases in humidity. **MQ′ 135 Gas Sensor:** The MQ′ 135 sensor detects harmful gases including carbon monoxide, ammonia, and benzene. Gas concentration levels above a predefined threshold may indicate gas leakage. **Sharp GP2Y1010AU0F Dust Sensor:** This optical sensor detects dust particles by measuring light scattered by airborne particles. It provides an estimate of particulate matter concentration. **PM2.5 Monitoring:** PM2.5 measurements help classify air quality conditions and detect pollution levels that may affect respiratory health. The sensor data is transmitted to a processing system where preprocessing techniques such as noise filtering, normalization, and anomaly detection are applied. Machine learning algorithms including Logistic Regression and Isolation Forest are used to analyze environmental patterns and detect abnormal conditions.

3. Machine Learning Analysis

Machine learning models play an important role in analyzing environmental sensor data. Logistic Regression is used as a classification model to determine whether environmental conditions are safe or unsafe based on sensor readings. Isolation Forest is used for anomaly detection to identify unusual environmental patterns that may indicate hazards such as gas leakage or abnormal temperature changes. These models allow the system to automatically learn patterns within environmental data rather than relying solely on static threshold values. This improves the reliability and accuracy of environmental hazard detection.

4. Results and Visualization

The system was tested using simulated environmental sensor readings. The visualizations below demonstrate relationships between environmental variables and how the monitoring system detects environmental changes. Correlation analysis reveals strong relationships between temperature, humidity, gas levels, and particulate matter concentrations. Time series analysis shows how sensor readings evolve over time as environmental conditions change. Scatter plots illustrate how humidity levels can indicate human presence inside the monitored environment. Finally, PM2.5 measurements are used to classify indoor air quality conditions.

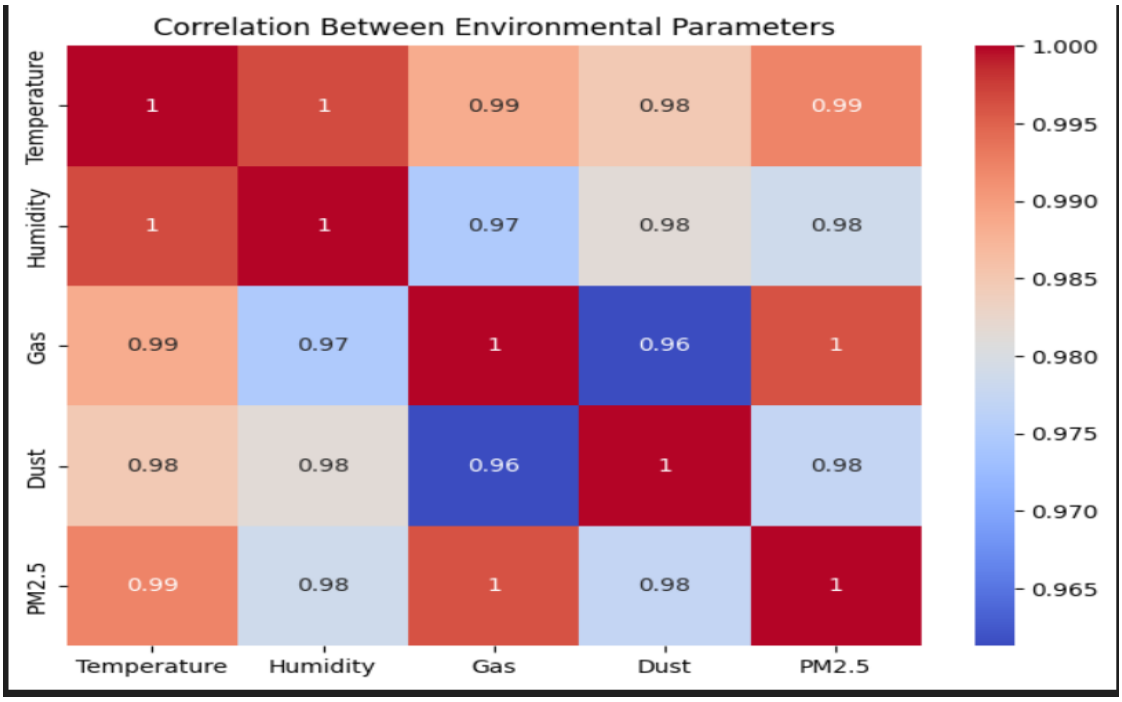

Figure 1: Correlation heatmap showing relationships between environmental parameters.

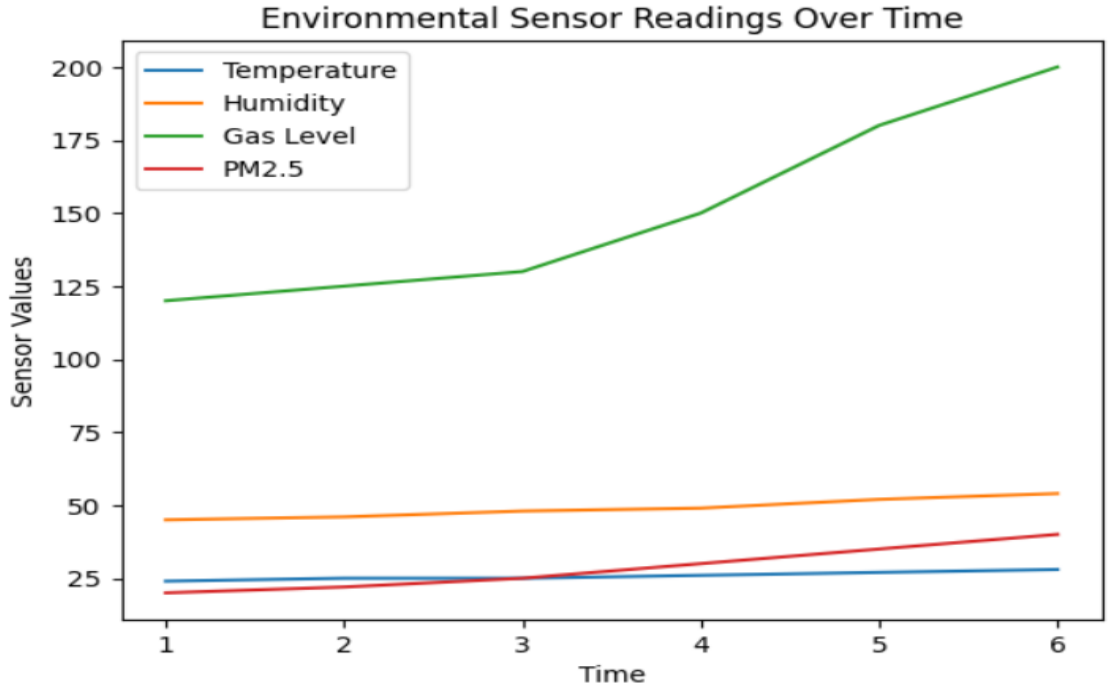

Figure 2: Environmental sensor readings over time.

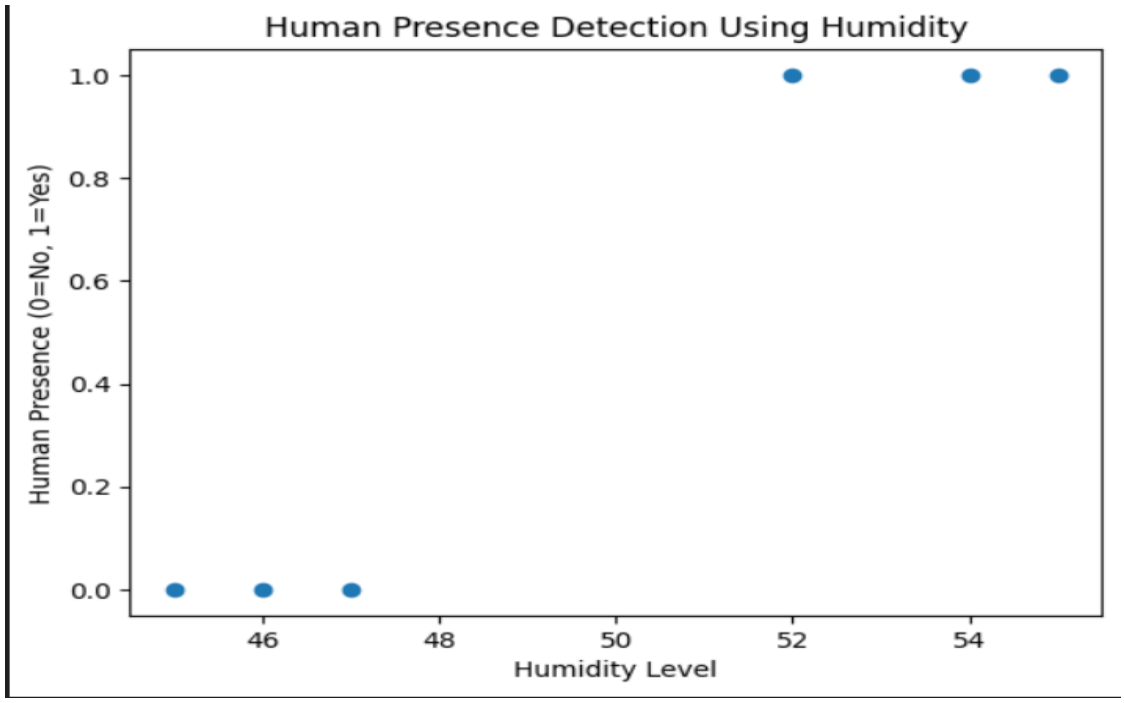

Figure 3: Human presence detection using humidity variations.

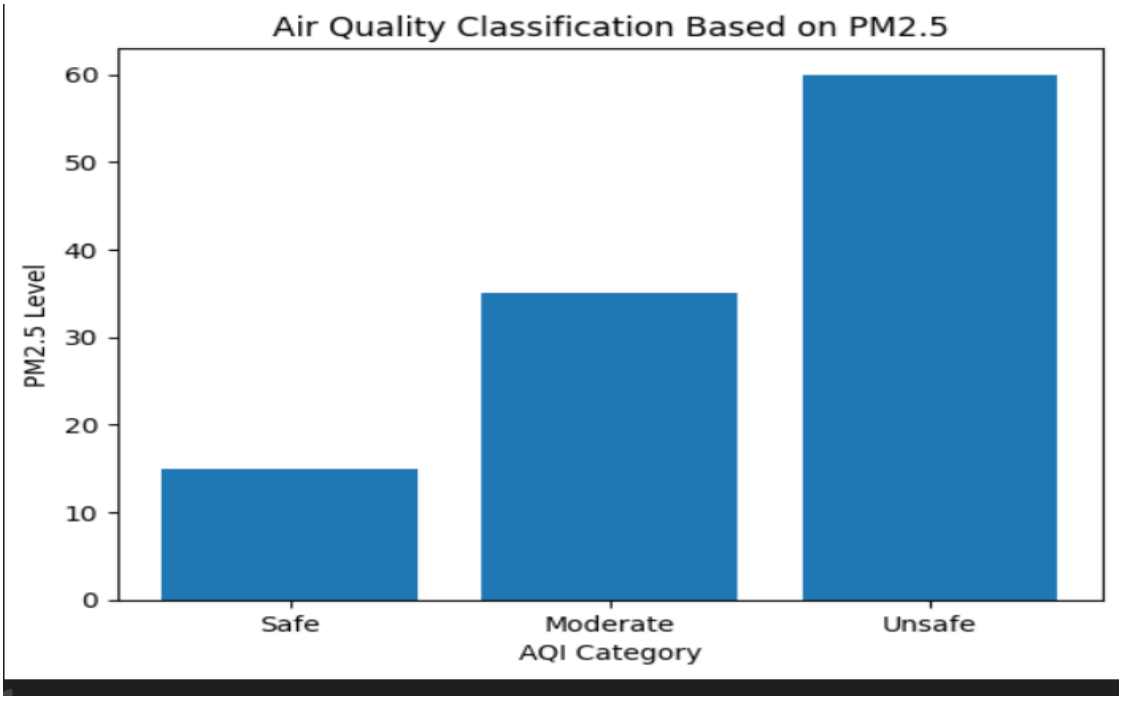

Figure 4: Air quality classification based on PM2.5 levels.

5. Conclusion

The proposed IoT-based indoor environmental monitoring system demonstrates how low-cost sensors combined with machine learning techniques can effectively detect hazardous environmental conditions. The integration of multiple sensors allows the system to monitor temperature, humidity, gas concentration, and particulate matter levels simultaneously. Machine learning models improve detection accuracy by identifying abnormal environmental patterns. The use of external air quality data obtained from the OpenAQ API provides additional context for analyzing environmental conditions. Future work may include integrating real-time alert systems, mobile monitoring applications, and advanced deep learning techniques for improved environmental prediction accuracy.